

STAT 516 MIDTERM; 50 pt total

1. (10pt) Tim has eight pairs of trousers, 15 shirts, six ties, and four jackets, out of which two pairs of trousers, five shirts, two ties, and two jackets are green. Suppose that one morning Tim selects his outfit completely at random. What is the probability that he chooses a completely green outfit?

Solution: The total number of possible ways that he could select his outfit is $8 \times 15 \times 6 \times 4 = 2880$. There are $2 \times 5 \times 2 \times 2 = 40$ possible ways that he could choose a completely green outfit. Thus, the probability is $\frac{40}{2880} = .014$

2. (15pt) Let X be the geometric random variable with probability of success equal to p at each trial. That is, the probability mass function of X is $p(x) = p(1-p)^{x-1}$ for $x = 1, 2, \dots$. Consider another random variable $Z = (X - 2)^2$. What is the probability mass function of Z ?

Solution: the range of values of Z is $D_Z = \{0, 1, 4, 9, 16, \dots\}$. Moreover, $Z = 0$ if and only if $X = 2$; $Z = 1$ if and only if $X = 1$ or $X = 3$; while for the other values of the range there is a one-to-one correspondence given by $x = \sqrt{z} + 2$ where $z \in \{4, 9, 16, \dots\}$. Hence, the pmf of Z is $p_Z(2) = p_X(2) = p(1-p)$ for $z = 0$; $p_X(1) + p_X(3) = p + p(1-p)^2$ for $z = 1$; and $p_X(\sqrt{z} + 2) = p(1-p)^{1+\sqrt{z}}$ for $z = 4, 9, 16, \dots$

3. (25pt) A fair die is rolled n times. What is the number of trials needed for the probability of at least one 6 to be greater than or equal to $\frac{1}{2}$?

Solution; the solution is given by the smallest integer n such that

$$1 - \left(\frac{5}{6}\right)^n \geq \frac{1}{2}$$

In other words, $n \geq -\log 2 * \log \frac{5}{6} \approx 0.126$. Thus, the smallest number of trials needed is 1.